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MILITARY SPECIFICATION

NOZZLE, PRESSURE FUEL SERVICING, LOCKING, TYPE D-1 AND D-2, NOMINAL 2 1/2 INCH DIAMETER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document specifies the requirements for two types of pressure fuel servicing nozzles, designated Types D-1 and D-2, which have a nominal inlet diameter of 2 1/2 inches.

1.2 Classification. Nozzles shall be of the following types, as specified (see 6.2).

TYPE D-1 - MS29520-1, 45° Inlet

TYPE D-2 - MS29520-2, Straight

Both types must employ a swivel fitting to facilitate hose handling during refueling.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1. Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Air Engineering Center, Systems Engineering and Standardization Department (93), Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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SPECIFICATIONS

FEDERAL

P-D-680	Dry Cleaning Solvent
TT-S-735	Standard Test Fluids; Hydrocarbon
UU-P-268	Paper, Kraft, Wrapping
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-C-1752	Cushioning Material, Packaging, Uncellular Polyethylene Foam, Flexible
PPP-C-1842	Cushioning Material, Plastic, Open Cell (for Packaging Applications)
PPP-E-540	Envelopes: Water-Resistant for Packing List and Shipping Documents

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-P-5315	Packing, Preformed, Hydrocarbon Fuel Resistant
MIL-C-5541	Chemical Conversion Coatings on Aluminum Alloys
MIL-G-5572	Gasoline, Aviation: Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aerospace Vehicles
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for
MIL-R-25988	Rubber, Fluorosilicone Elastomer, Oil and Fuel Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes
MIL-T-38219	Turbine Fuel, Low Volatility, JP-7

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MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-R-83248	Rubber, Fluorocarbon Elastomer, High Temperature Fluid and Compression Set Resistant
MIL-C-83488	Coating, Aluminum Vapor Deposited

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking for U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1523	Age Control of Age Sensitive Elastomeric Material
MIL-STD-1629	Procedures for Performing a Failure Mode, Effects and Criticality Analysis
MIL-STD-2175	Casting, Classification and Inspection of
AN763	Gasket - Swivel, Fitting Flanged
MS24484	Adapter, Pressure Fuel Servicing, Aircraft Nominal 2 1/2 Inch Diameter
MS26551	Strainer, 2 1/2 Inch Pressure Fuel Servicing
MS29513	Packing, Preformed, Hydrocarbon Fuel Resistant, "O" Ring
MS29520	Envelope Dimensions, Nozzle, Pressure Fuel Servicing Locking Aircraft, Type D-1 and D-2
MS33666	Packing, Preformed - Aeronautical, Elastomeric, Range of Sizes
MS33786	Fitting Installation, Flared Tube and Hose, Swivel

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Commercial Packaging, Practice for (DOD ADOPTED)

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The D-1 and D-2 nozzles furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Materials. Materials and processes used in the manufacture of nozzles shall be resistant to all fluids as specified in TT-S-735, MIL-G-5572, MIL-T-5624, MIL-T-38219, and MIL-T-83133. The use of copper, cadmium, magnesium, or any alloy thereof is prohibited.

3.2.1 Metals. Corrosion resistant metals shall be used. Exterior surfaces susceptible to contact with ferrous surfaces during use shall be nonferrous to restrict spark generation.

3.2.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined by MIL-STD-889 shall not be used in contact with each other.

3.2.1.2 Protective coatings. All aluminum alloy parts shall be protected by coatings in accordance with MIL-A-8625, MIL-C-5541, MIL-C-83488, or other protective coatings which will withstand the effects of this specification.

3.2.1.3 Color Identification. The nozzle shall employ a red bumper or red embossed nameplate to identify use with fuel.

3.2.2 Castings. Castings shall be classified and inspected as specified in MIL-STD-2175.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.4 Design. The nozzle shall be designed to operate within the clearance envelope of MS29520 and shall mate with the MS24484 adapter within a misalignment angle up to 4 degrees. The nozzle shall transfer fuel at flows up to 800 gallons per minute (gpm) with operating pressures of up to 55 psi. The fuel flow shall be controlled by an ON/OFF fuel flow actuation lever separate from the handles used to hold or attach the nozzle to the adapter. Actuation of the lever shall simultaneously open or close the adapter sealing unit and permit flow to or from the aircraft. Operation of the lever from the nonflow to flow position shall be impossible unless the nozzle is secured onto the adapter. Removal of the nozzle from the adapter shall be impossible unless the lever is in the nonflow position. The lever shall remain in either the full flow or nonflow position under the full range of flows and pressures, during fueling or defueling, without the need of maintaining continuous manual holding force. To prevent leakage, the nozzle-to-adapter main seal shall automatically conform to the adapter when the nozzle is attached and opened. The seal must compensate for 0.020 inch wear on the adapter lugs, dimensional tolerances, side loading, and temperature changes. Provision should be made for the installation and retention of a screen conforming to MS26551. The inlet flange shall be designed to accommodate an AN763-40 gasket.

3.4.1 Swivel inlet. The nozzle shall employ a swivel inlet fitting capable of 360° rotation to allow the hose connection to turn without hinderance.

3.4.2 Dust cap. A nonmetallic cap shall be provided to prevent dust from entering that portion of the nozzle which engages the adapter and protect the seal face of the nozzle during handling when tested to the requirements of MIL-STD-810, (see 4.6.4). When not in use, the cap shall hang free outside the refueling envelope so that it will not interfere with the refueling. Retention springs are not permitted.

3.4.3 Lubricants. The nozzle shall operate satisfactorily without the use of lubricants.

3.4.4 "O" ring packing and seals. "O" ring packing and associated seals shall conform to MIL-P-5315, MIL-R-25988, MIL-R-83248, MS29513, or MS33666 as applicable, or other materials approved by qualification testing (see 4.3.1).

3.4.4.1 Age controls for synthetic rubber parts. Age sensitive synthetic components shall be marked in accordance with MIL-STD-1523.

3.4.5 Maintenance. The nozzle shall not require special tools or fixtures for overhaul or repair of the assembly or any of its components.

3.4.6 Pressure drop. The pressure drop through the nozzle and the MS24484 adapter shall be not greater than 10 pounds per square inch (psi) corrected when subjected to a 600 gpm flow of fuel with no MS26551 strainer installed. In the defueling mode, the pressure drop through the nozzle and adapter shall be not greater than 2 psi corrected when subjected to a 200 gpm flow of fuel with the strainer removed. The test setup shall be in accordance with Figure 1.

3.4.7 Flow-actuation lever. The flow-actuation lever shall be capable of operating from full closed to full open with a maximum torque of 40 pounds force inch (lbf•in) when a force of 50 pounds is applied against the nozzle poppet in the direction to close the nozzle.

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3.4.8 Strength. The nozzle shall be able to withstand the loads shown on Figures 2 and 3 without leakage, distortion, or failure when tested in accordance with 4.6.6.

3.4.9 Fuel quality sampling port. The nozzle shall be provided with a sampling port boss having a .375 inch National Pipe Thread (NPT). Location of the boss shall allow insertion of a sampling probe which can penetrate 0.75 inch within the nozzle flow stream. Insertion of the probe shall not interfere with the operation of the nozzle or the MS26551 screen.

3.4.10 Electrical continuity. Electrical continuity between the nozzle and adapter shall be established through the attaching lugs when the units are connected. The resistance from the nozzle inlet flange to the adapter locking flange shall be not greater than 10 ohms.

3.5 Construction. The nozzle shall be so constructed that it will pass the tests defined in Section 4.

3.6 Performance. The nozzle shall satisfy all performance requirements when subjected to any tests of 4.2 in the order listed.

3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable in accordance with MIL-I-8500. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Weight. The weight of the nozzle shall be not greater than 9 pounds.

3.9 Threads. All machine screw threads shall be in accordance with MIL-S-7742 or MIL-S-8879.

3.9.1 Locking of threaded parts. All threaded parts shall be locked to prevent loosening or detachment which would result in the part entering the system being refueled or cause foreign object damage to the aircraft.

3.10 Operational/Maintenance Manual. A manual which provides specific instructions for the operation, maintenance, and repair of the nozzle by refueling personnel shall be provided and approved in accordance with 4.3.4 and 5.1.1.2.

3.11 Identification of product. The nozzle shall be marked for identification in accordance with MIL-STD-130. The identification shall be permanent and its location shall not affect nozzle performance. The minimum information is as follows:

Nozzle, Pressure Fuel Servicing
Contract No.
Type D-(insert applicable dash number)
Manufacturer's identification, part number, serial number
Cure date per MIL-STD-1523 (if applicable)

3.12 Safety. The nozzle shall be designed to minimize fuel sprays or spills resulting from failure of any nozzle component. The manufacturer shall prepare a detailed nozzle Failure Modes and Effects Analysis (FMEA) in accordance with MIL-STD 1629, which will at a minimum, list each individual

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component or piece part, its purpose, each possible component failure; i.e., loose, broken, missing, etc., and the effect of each failure under operating conditions. The government will determine any corrective actions deemed necessary by the FMEA.

3.13 Workmanship. The nozzle shall be uniform in quality, clean and free from faults and burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classifications of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall consist of qualification tests on two nozzles by the contractor as stipulated in para 4.3.1, and verification tests on eight additional untested nozzles at a Government facility. The verification tests shall consist of any of the tests of 4.6 which the Government desires to perform. Successful completion of this inspection is required for qualification. Samples submitted shall be accompanied by two complete sets of detail and assembly drawings and a complete test report. The detail and assembly drawings are supplied for qualification and conformance only, and for no other purpose. Samples shall be submitted to the Naval Air Propulsion Center, P.O. Box 7176, Trenton, NJ 08628-0176.

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4.3.1 Qualification tests. The qualification tests shall consist of the following:

Sample No. 1

(1)	Examination of product	(4.6.1)
(2)	Proof pressure	(4.6.9)
(3)	Calibration	(4.6.2)
(4)	Dust test	(4.6.4)
(5)	Swivel endurance	(4.6.5)
(6)	Endurance	(4.6.7)
(7)	Electrical conductivity	(4.6.12)
(8)	Disassembly and inspection	(4.6.13)

Sample No. 2

(1)	Examination of product	(4.6.1)
(2)	Calibration	(4.6.2)
(3)	Fuel resistance and low temperature	(4.6.3)
(4)	Salt water immersion	(4.6.8)
(5)	Strength	(4.6.6)
(6)	Leakage	(4.6.2.3)
(7)	Burst pressure	(4.6.10)
(8)	Drop test	(4.6.11)
(9)	Disassembly and inspection	(4.6.13)

4.3.1.1 Rejection and retest. When a nozzle fails to meet any of the requirements of the qualification tests, the program will be halted until the extent and cause of failure are determined. Prior to reinitiation of the test program, full particulars of the failure and the action required to correct the fault will be reported to the qualifying activity. All qualification tests which are affected by the corrections shall be repeated.

4.3.2 Manufacturer's drawings. Included with the manufacturer's drawings submitted with the qualification test specimens shall be a cutaway section showing all parts in their normal assembled positions and shall specify part

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numbers of all parts and subassemblies. The drawings shall be considered proprietary to the vendor. In addition, the following information shall be furnished on or together with the assembly drawings.

a. Envelope dimensions and mounting provisions.

b. Materials, construction, treatment, and finish.

c. Pressure and flow rate ratings.

4.3.3 Test report. The test report submitted with the untested qualification test specimens shall conform to MIL-STD-831 and shall include the following:

a. Report of all tests, together with a detailed statement indicating conformance or extent of conformance with all requirements of this specification, referring specifically to paragraph numbers. Wherever a requirement is considered to be not applicable, the report should so state.

b. Summary of endurance test.

c. Diagrams of all test setups.

d. Outline and description of tests and test conditions.

e. Copies of test log sheets.

f. Photographs.

g. List and descriptions of testing equipment.

h. List of where specific tests were conducted.

i. The Failure Modes & Effects Analysis (FMEA) for the nozzle.

4.3.4 Operational/Maintenance Manual. The manual shall be provided with the test samples detailing the recommended operation, maintenance, and repair of the nozzle. Format for the manual shall be in accordance with normal commercial practice. Review and approval of the manual by the qualifying activity is necessary prior to release for procurement purposes (see 3.10 and 5.1.1.2).

4.3.5 Retention of qualification. To retain qualified status, each manufacturer is required, at intervals not exceeding two years, to forward to the qualifying activity, a certificate signed by a responsible official of management. The certification shall state that all conditions are the same that existed at the time of qualification and that the requirements of the current specification can be met.

4.4 Quality conformance inspection. Quality conformance inspection shall be performed under the surveillance of the Government Inspector on lots submitted for acceptance under contract. The quality conformance test shall consist of individual tests and sampling tests.

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4.4.1 Individual tests. Each nozzle shall be subjected to the following tests, as described under 4.6.

a. Examination of product (see 4.6.1).

b. Functional, no flow required (see 4.6.2.1).

c. Leakage, +10 and 60 psig only (see 4.6.2.3).

4.4.2 Sampling plan and tests.

4.4.2.1 Sampling plan. One nozzle shall be selected at random from each production lot up to 200 units, and one out of every 500 units thereafter.

4.4.2.2 Sampling tests. The nozzles, selected in accordance with 4.4.2.1 shall be subjected to the following tests, which shall be in addition to the individual tests specified in 4.4.1.

a. Calibration (see 4.6.2).

b. Proof pressure (see 4.6.9).

c. Disassembly and inspection (see 4.6.13).

4.4.2.3 Failures. When an item selected from a production run fails to conform to the specification, no items still on hand or produced later shall be accepted until the extent and cause of failure have been determined, reported on, and appropriately corrected. Reports shall be forwarded in accordance with 4.3.3.

4.5 Test conditions. Unless otherwise specified, the following test conditions shall apply.

4.5.1 Test fluid. Fluid in accordance with MIL-G-5572, grades 100/130 or 115/145; MIL-T-5624, grades JP-4 or JP-5, MIL-T-83133, grade JP-8, or TT-S-735, type I and type III, shall be used for testing. Fluid conforming to P-D-680, type II, or any fluid acceptable to the contracting activity, may be used when a specific fluid is not specified.

4.5.2 Ambient temperature and pressure. All tests shall be performed with the ambient and test fluid at temperatures of 60°F to 90°F and at local barometric pressure.

4.5.3 Test adapter. The test adapter used shall be an MS24484 installed in a test housing (see Figure 4).

4.6 Test methods.

4.6.1 Examination of product. Each nozzle shall be subjected to inspection for adherence to the envelope requirements of the MS29520 and applicable design drawings. The nozzle shall be examined to determine conformance to the requirements of this specification with respect to materials, interchangeability, construction, finish, identification, workmanship, weight, location of sampling port, and protective coatings. The foregoing shall be noted and recorded.

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4.6.2 Calibration.

4.6.2.1 Functional. The nozzle shall be inserted and attached onto the test adapter (see Figure 4) and the adapter sealing valve actuated a minimum of five times. Test fluid shall be passed through the nozzle and adapter with flows from 100 to 800 gpm in 100 gpm increments with a minimum inlet static pressure of 40 psig under flow conditions and 55 psig in the nonflow position. The adapter sealing valve shall be actuated to open and shut a minimum of three times at each flow condition, and there shall be no evidence of binding or chattering of the lever. The torque required to actuate the lever shall be less than 40 lbf•in. under no-flow conditions.

4.6.2.2 Pressure drop. Using a test setup similar to Figure 1, and the MS24484 adapter and test housing in accordance with Figure 4, a pressure drop test shall be performed with flows from 0 to 800 gpm, in 100 gpm increments, with an inlet pressure of 40 psi. Test results shall reflect both actual and corrected pressure drops. The formula for correction is as follows:

$$\Delta P = \Delta P_{NA} \times \left(\frac{1.34}{\mu}\right)^{.25}_{cs} \times \left(\frac{.769}{\rho}\right)$$

where μ = kinematic viscosity of test fluid

ρ = specific gravity of test fluid

4.6.2.3 Leakage.

4.6.2.3.1 Connected. With the nozzle connected to the adapter (see Figure 4) and the nozzle flow-actuation lever in the full open position, the leakage test shall be conducted over a range of -12 to 4 psig in 2 psi increments and from 10 to 60 psig in 10 psi increments. The pressure shall be maintained at each increment for at least one minute. There shall be no evidence of external fuel leakage from the nozzle, and air leakage inward into the nozzle shall not exceed 50 cc per minute during any part of the leakage tests.

4.6.2.3.2 Disconnected. With the nozzle disconnected from the adapter, fuel pressures of 0 to 60 psig in 10 psi increments, air at pressures from -12 to 0 psig in 2 psi increments, shall be applied to the nozzle inlet. Pressure shall be maintained for at least one minute at each increment. There shall be no external fuel leakage nor air leakage into the nozzle.

4.6.2.4 High flow shutoff. With the nozzle connected to the adapter (Figure 4) and test housing, conduct ten rapid shutoffs with the flow-actuation lever while flowing 400 gpm and 600 gpm at an inlet pressure of 60 psig. There shall be no evidence of deformation, fracture, or malfunction of the nozzle.

4.6.3 Fuel resistance and low temperature. The fuel resistance and low temperature tests shall be conducted in accordance with Table I.

4.6.4 Dust. The nozzle, with the dust cover in place and the inlet sealed, shall be subjected to test method 510.2 of MIL-STD-810. The functional test requirement of 4.6.2.1 shall be conducted following completion of the dust test.

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4.6.5 Swivel endurance. The nozzle swivel shall be wear-tested by loading the nozzle with a 100 lbf•ft bending load (see Figure 2) and turning the swivel 2000 times through at least a 45° arc. During this test, the nozzle will be pressurized with test fluid to 55 psig. Following this loading, the nozzle shall successfully complete the proof pressure test of 4.6.9.

4.6.6 Strength. The MS24484 adapter shall be retained by the mounting flange only, to a fluid container capable of sustaining any pressures applied. The nozzle shall be connected and the adapter valve actuated to the flow position. The nozzle test adapter shall be filled with test fluid. There shall be no evidence of leakage, deterioration, deformation, fracture, or malfunction when the following loads are applied to the nozzle (Figures 2 and 3).

a. 25 applications of a 600 pound force foot (lbf•ft) bending load, applied as shown on Figure 2.

b. 950 applications of a 350 lbf•ft bending load, applied as shown on Figure 2.

c. 25 applications of a 200 lbf•ft bending load, applied as shown on Figure 2, with a fluid pressure of 240 psig applied to the adapter-nozzle combination.

d. 100 applications of a combined load of a 100 lbf•ft couple applied to the nozzle handles and an axial load of 200 pounds applied to the nozzle inlet flange in accordance with Figure 3, with a fluid pressure of 240 psig applied to the adapter-nozzle combination.

4.6.7 Endurance. There shall be no evidence of binding, chatter, or leakage during the tests of 4.6.7.1 and 4.6.7.2. The force to actuate the lever shall not exceed the requirements of 4.6.2.1.

4.6.7.1 Normal operation. The nozzle shall be operated for 2,000 cycles (6.4.1) with fuel pressure of 55 psig applied to the nozzle in the closed position and with fuel flow through the nozzle in the open position at a minimum rate of 400 gpm.

4.6.7.2 Maximum loading. The nozzle, filled with fuel, shall be connected to the MS24484 adapter in a test setup capable of imposing a 25 $\pm^{3}_{0}$ pounds load on the poppet in the closed position in a direction tending to close the valve. After the valve is opened 0.75 inch, the load shall be increased to 50 $\pm^{5}_{0}$ pounds. The valve shall be fully opened and closed 2,000 times. Upon completion of this test, the leakage test (4.6.2.3) shall be conducted at a positive pressure only.

4.6.8 Salt water immersion. The nozzle shall be immersed in a solution consisting of 5 percent by weight of NaCl in distilled water. After immersion, the solution shall be drained and the nozzle shall be heated in an oven to a temperature of 130 ±5°F for a period of not less than one hour. The immersion and heating cycle shall be repeated 50 times. The nozzle shall not be operated at any time during the immersion cycles. Immediately after completing the immersion cycles, the nozzle shall be washed with warm water to remove all salt accumulations. The nozzle shall be dried, wetted with fuel, and operated for three complete cycles (see 6.4.1). Corrosion of any part of the nozzle to a degree that affects performance shall be cause for rejection.

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4.6.9 Proof pressure. With the nozzle in the closed position, it shall be subjected to a pressure which shall be not less than 180 psig for at least one minute. There shall be no evidence of leakage, distortion, or other damage to any part of the assembly.

4.6.10 Burst pressure. With the nozzle in the closed position, it shall be subjected to a pressure which shall be not less than 240 psig for at least one minute. There shall be no evidence of distortion or other damage to any part of the nozzle; and when the pressure is lowered to 60 psig, there shall be no evidence of leakage from any portion of the nozzle.

4.6.11 Drop test. The nozzle, attached to a 2 1/2 inch ground refueling hose, shall be dropped at least ten times from a height of 6 feet onto a steel plate. Drops shall be configured to impact at least once upon the handles, lever, and the nozzle seal face. The remaining impact shall be at random locations as specified by the Naval Air Propulsion Center. The steel plate shall be at least 1/2 inch thick resting on a concrete floor. The upstream portion of the hose shall lie at floor level during the test. Upon completion of this test, the nozzle shall satisfactorily pass the functional (4.6.2.1) and leakage (4.6.2.3) tests.

4.6.12 Electrical conductivity. With the nozzle attached to the MS24484 adapter, a resistance reading shall be made between the nozzle inlet and the adapter mounting flange. Resistance shall be not greater than 10 ohms.

4.6.13 Disassembly and inspection. The nozzle shall be disassembled for inspection of all parts. All anomalies shall be photographed and discussed in the test report. There shall be no defects that would cause the nozzles to fail to function within specification limits.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Level A preservation shall be Method III in accordance with MIL-P-116.

5.1.1.1 Cleaning and drying. The nozzle shall be cleaned and dried by a process or combination of processes which will accomplish a thorough cleaning and drying without damage to the nozzle in accordance with MIL-P-116.

5.1.1.2 Unit pack. The unit pack shall contain one nozzle, cleaned as in 5.1.1.1, one AN763-40 gasket and one manufacturer's maintenance manual. The nozzle shall be placed in a container conforming to type CF class weather resistant of PPP-B-636 and if necessary, cushioned with material conforming to PPP-C-1842 or PPP-C-1752. The gasket shall be wrapped in Kraft paper conforming to UU-P-268. The manual shall be placed in an envelope conforming to PPP-E-540, class 4.

5.1.2 Commercial. The nozzle shall be preserved in accordance with ASTM D 3951.

5.2. Packing. The shipping containers shall be of minimum cube consistent with the level of protection specified (see 6.2). All shipping containers will contain the same number of unit packs snugly packed. The levels of packing shall be A, B, or Commercial.

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5.2.1 Level A. The specified number of unit packs shall be packed in overseas containers conforming to PPP-B-621 or PPP-B-601 (see 6.2). Strapping and closure shall be in accordance with the appendix of the applicable shipping container.

5.2.2 Level B. The specified number of unit packs shall be packed in weather resistant containers conforming to PPP-B-636 or PPP-B-640 (see 6.2). Closures shall be in accordance with the appendix of the applicable container specification.

5.2.3 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Each unit pack and shipping container shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The nozzles covered by this specification are intended for use in refueling or defueling an aircraft by means of a pressure fuel servicing system. The type D-1 nozzle incorporates a 45° inlet configuration for preferential horizontal refueling. The type D-2 nozzle incorporates a straight inlet configuration for preferential vertical refueling.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type (D-1 or D-2) nozzle (see 1.2 and 6.1).

c. Levels of preservation, packaging, and packing (see Section 5).

6.3. Qualification. With respect to products requiring qualification, awards will be made for such products which are, prior at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List (QPL), whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Engineering Center, SESD (Code 9311), Lakehurst, NJ 08733-5100. However, information pertaining to qualification of products may be obtained from the Naval Air Propulsion Center, P.O. Box 7176, Trenton, NJ 08628-0176.

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6.4 Definitions.

6.4.1 Cycle. Unless otherwise specified, a cycle is defined as connecting the nozzle to the test adapter, fully opening and closing the flow-actuation lever four times, and removing the nozzle from the test adapter.

6.5 Subject term (key word) listing.

Aircraft pressure defueling
Aircraft pressure refueling
Fueling
Nozzle

6.6 International standardization agreements. Certain provisions of this specification (3.4 and 4.6.5.1) are the subject of international standardization agreement ASCC AIR STD 25/4 and STANAG 3105. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.7 Changes from the previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:

Navy - AS
(Project No. 4930-0279)

Custodians:
Air Force - 99

Review activities:
Air Force - 82
DCSC-SS

User activities:
Army - AV, ME

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TABLE I. Fuel resistance and low temperature test schedule.

Test Period 1/	Fuel resistance				Low Temperature
	Phase I, Soak	Phase I, Dry	Phase II, Soak	Phase II, Dry	
Component configuration	Valve closed during first 72 hours and open during last 24 hours 2/	Drained, valve closed, inlet port open	Valve closed during first 12 hours and open during last 6 hours 2/	Drained, valve closed, inlet port open	Valve closed 2/
Test fluid	TT-S-735 type III	None	TT-S-735 type III	None	TT-S-735 type I
Period duration	96 hours (4 days)	24 hours	18 hours	30 hours	18 hours
Ambient and test fluid temperature	158° ± 2°F	Circulating air at 158° ± 2°F	158° ± 2°F	Circulating air at 158° ± 2°F	Lower the fluid temperature to -35 ± 2°F, then maintain the fluid temperature at -35 ± 2°F for a minimum of 18 hours
Operation or tests during period	Connect and disconnect nozzle and actuate valve once every 6 hrs.	None	Connect and disconnect nozzle and actuate valve once every 6 hrs.	None	None

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TABLE I. Fuel resistance and low temperature test schedule - Continued

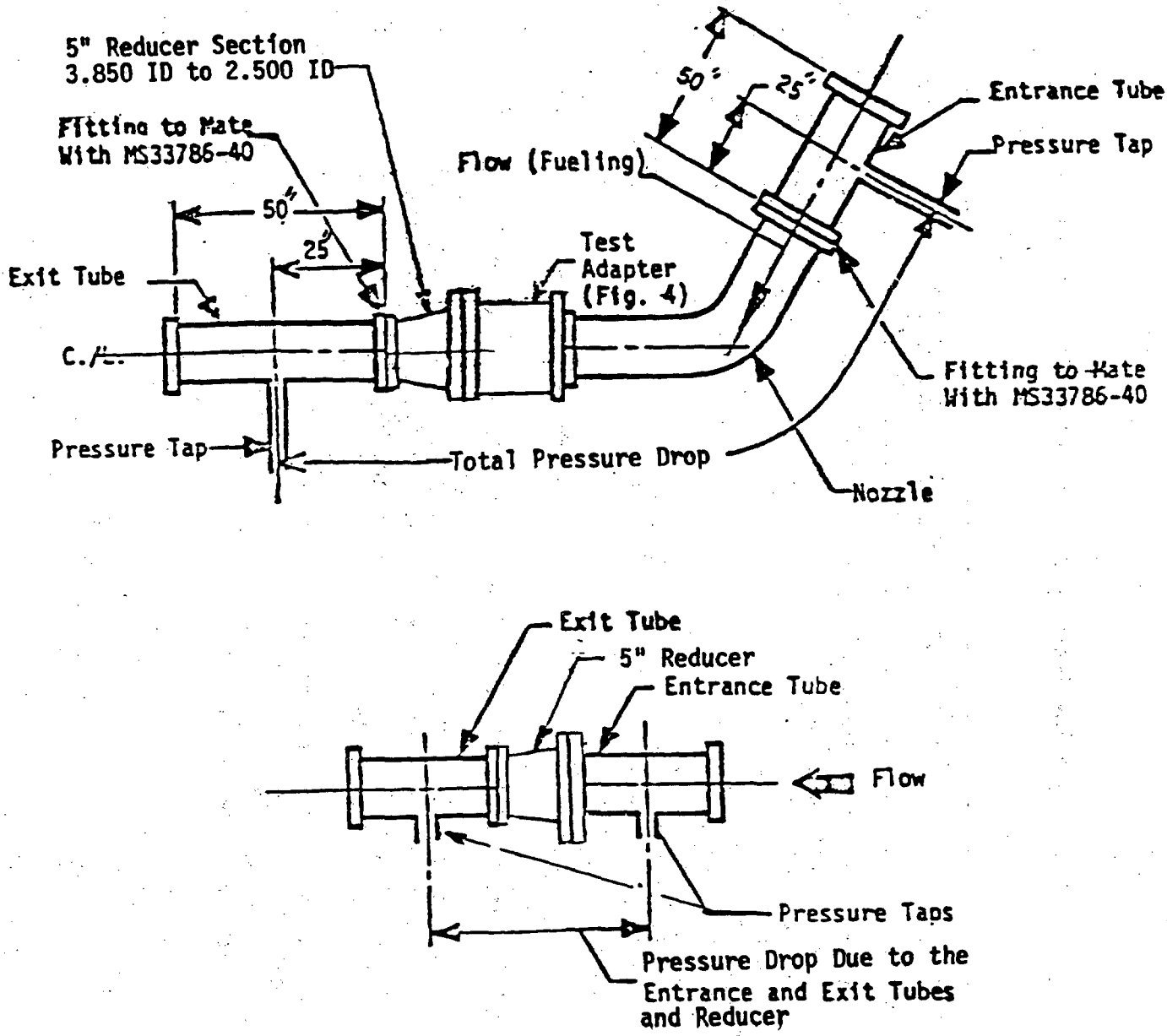
Test Period 1/	Fuel resistance				Low Temperature
	Phase I, Soak	Phase I, Dry	Phase III, Soak	Phase II, Dry	
Operation or tests immediately after period	Conduct leakage test (4.6.2.3), using TT-S-735 type III fluid at positive pressures only	Connect and disconnect nozzle and actuate valve once in dry condition. Conduct leakage test (4.6.2.3), using TT-S-735 type I fluid at positive pressures only	Same as phase I soak	Same as phase I dry	With ambient temperature and test fluid at -35°F: (a) operate the nozzle through 10 complete cycles. (b) Conduct leakage test, using TT-S-735 type I fluid at positive pressures only. 3/

1/ Each period shall follow immediately after the preceding one in the order noted.

2/ The component shall be maintained in such a manner as to insure complete contact of all non-metallic parts with the fluid as would be expected under normal service conditions. An assembly test housing conforming to Figure 4 shall be used when the valve is open. No connection shall be made during soak period when the nozzle is closed, except for test purposes.

3/ For the low temperature test (4.6.3), the leakage limits are:
Static: 0-4 psig - No leakage
Static: 4-60 psig - 2 ml per minute
During actuation: 0-60 psig - 2 ml per actuation

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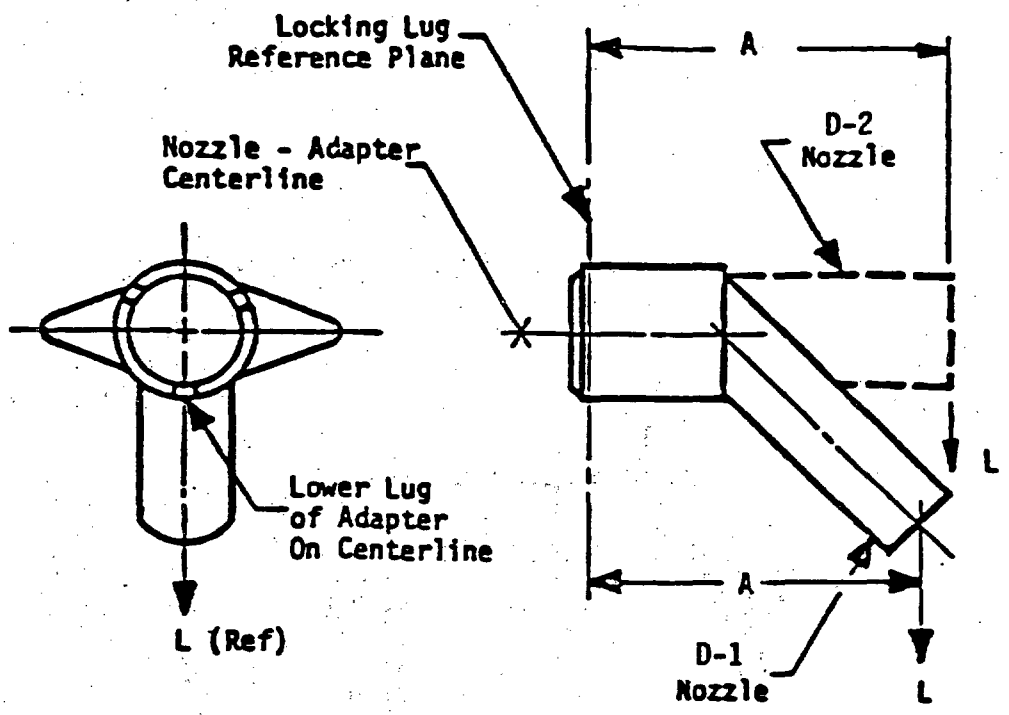


P_{NA} = Total pressure drop - pressure drop due to piping.

Nominal tube size - Entrance and exit tube - 2 1/2" Schedule 40

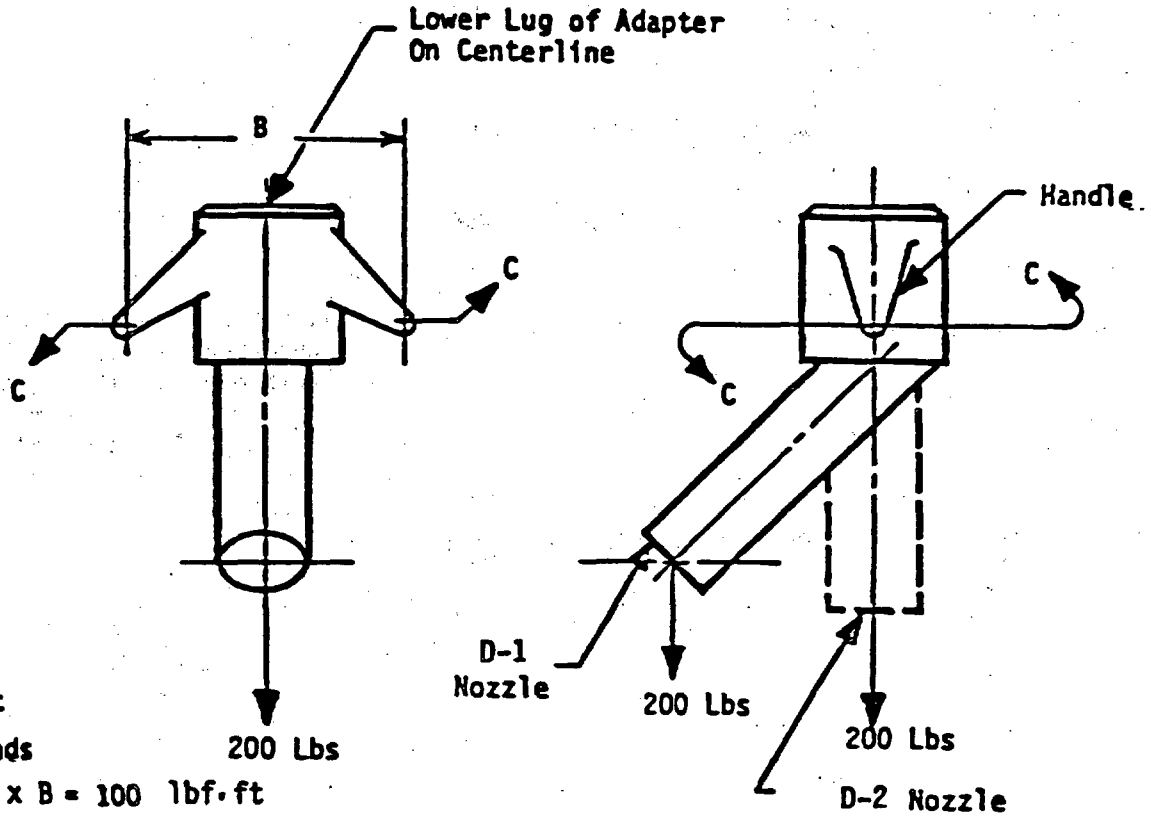
FIGURE 1. Pressure drop test.

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"A" in Feet
"L" in Pounds
Bending Load = L x A

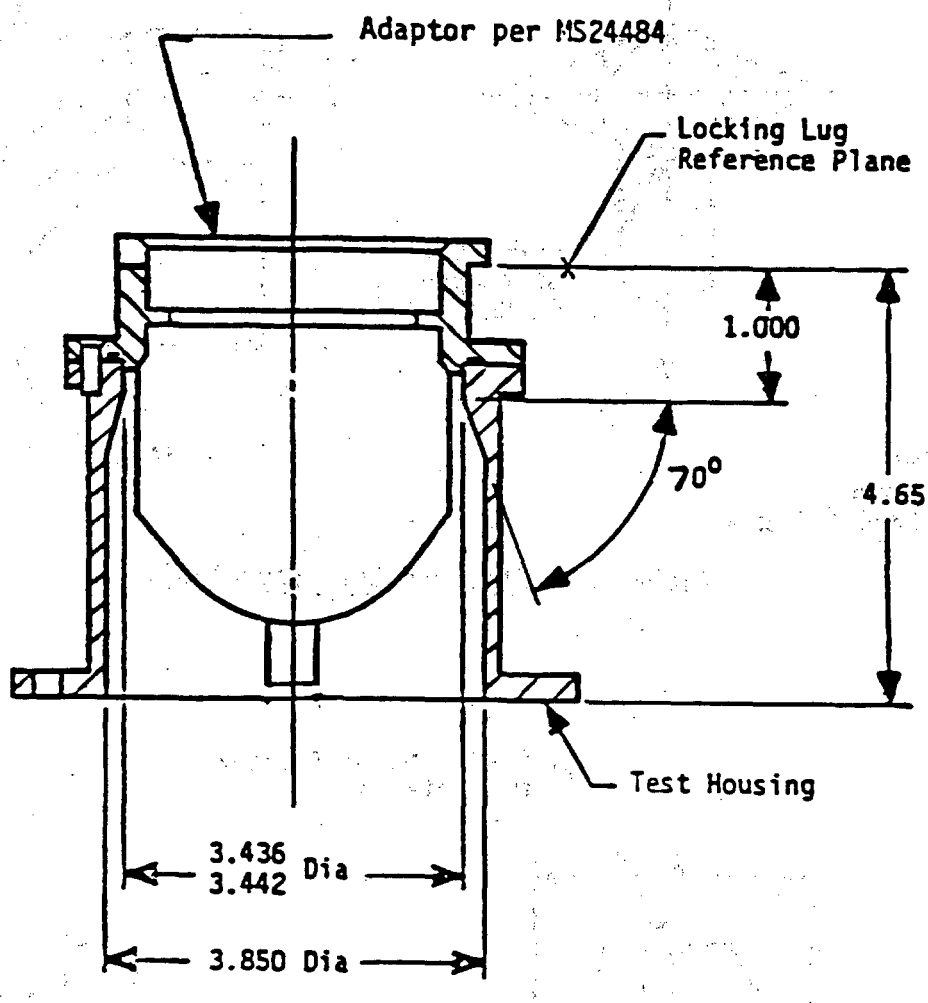
FIGURE 2. Strength test (bending).



"B" in Feet
"C" in Pounds
Couple = C x B = 100 lbf·ft

FIGURE 3. Strength test (combined loading).

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UNLESS OTHERWISE SPECIFIED
Dimensions are in inches.
Tolerances: .XX = ±.03
.XXX = ±.010
Angles = ±1.0°

FIGURE 4. Test housing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-N-5877D	Nozzle, Pressure Fuel Servicing, Locking, Type D-1 and D-2, Nominal 2 1/2 Inch Diameter

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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